INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, BHOPAL
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Parkinson's Disease Gait Severity Classification Using Sensor-Based Data



Chaiya Dhandhara – 24U020008
Atul Joshi – 24U020067

Dr. Meenakshi Choudhary
Assistant Professor
Department of CSE

Introduction

Understanding gait in Parkinson's disease through sensor-based analysis

Parkinson's disease is a progressive neurological disorder affecting movement, coordination, and balance

Gait changes include shorter steps, shuffling movements, reduced arm swing, and changes in walking rhythm

Sensor-based movement recordings provide quantitative information about motor impairment

Multichannel gait measurements analysed using machine learning and deep learning

PROJECT AIM

Classify Parkinson's disease gait severity into three categories using sensor-based movement data.

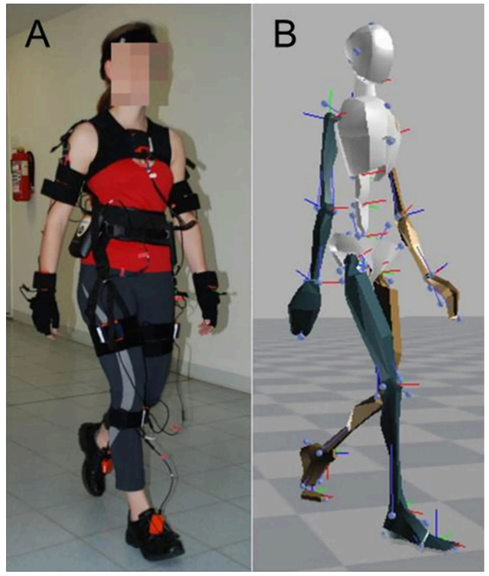
DATASET OVERVIEW

The dataset used in your project was collected and published by Shida in 2023.

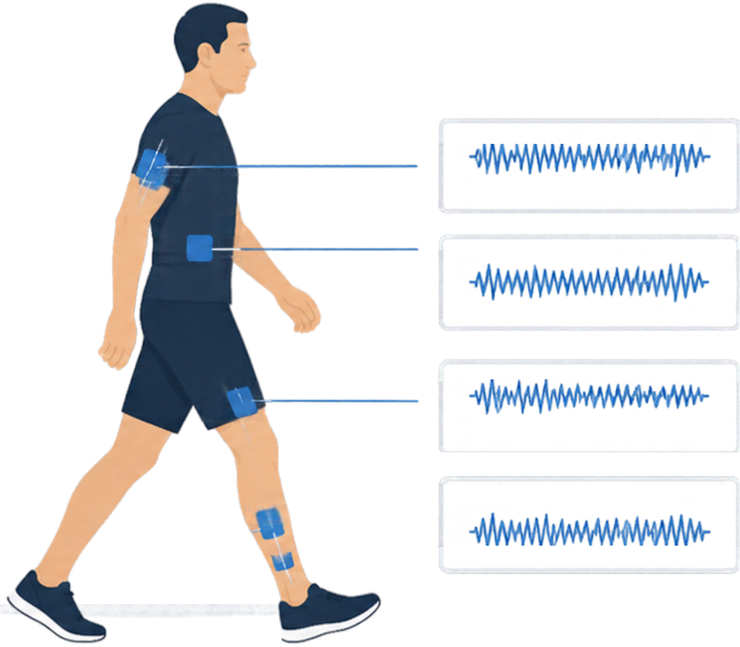
26 individuals with Parkinson's disease.

ON- and OFF-medication recordings where available.

Clinical walking scores used as target labels.



Source:frontiersin.org



Background

Sensor-based gait analysis provides **quantitative walking pattern information.**

Clinical assessment uses Movement disorder society – Unified Parkinson's disease rating scale (**MDS-UPDRS)** for motor symptom evaluation.

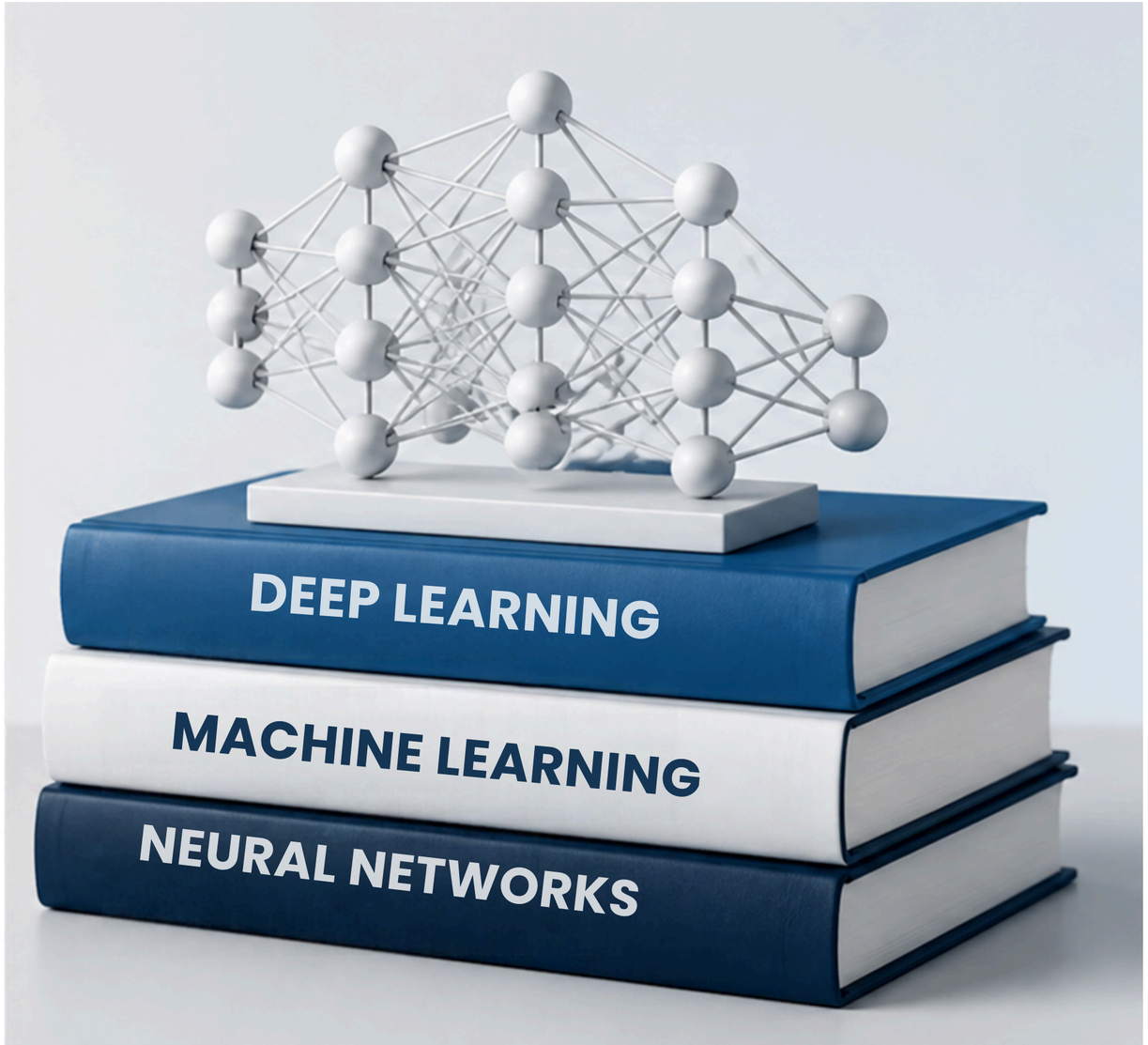
Machine learning has been explored for Parkinson's disease gait analysis.

Deep learning enables automatic feature learning from sequential data.

RESEARCH GAP

Underutilization of sensor data

- Previous studies often rely on skeletal joint trajectories or manually extracted gait features.



Problem Statement



Problem

Gait recordings contain multiple measurements collected over time, making manual analysis difficult. Therefore, we aim to develop a computational framework that can learn movement patterns from these recordings and classify gait severity into three predefined categories

Challenges

- Multichannel sequential gait data
- Data preparation and quality issues
- Different models capture different movement patterns
- Trial-wise evaluation may include same subjects across partitions

Objectives

- Explore and validate dataset
- Integrate linear and angular measurements
- Prepare consistent movement sequences
- Implement and compare five classification models
- Evaluate using accuracy, balanced accuracy, and macro F1-score

Proposed Methodology

01 Dataset Exploration

Parkinson's gait dataset.

26 participants.

ON/OFF conditions.

02 Data Preparation

Linear and angular data integration.

Data quality assessment.

Sequence preparation.

Feature scaling.

03 Model Development

CNN1D

LSTM

BILSTM

BILSTM-CNN

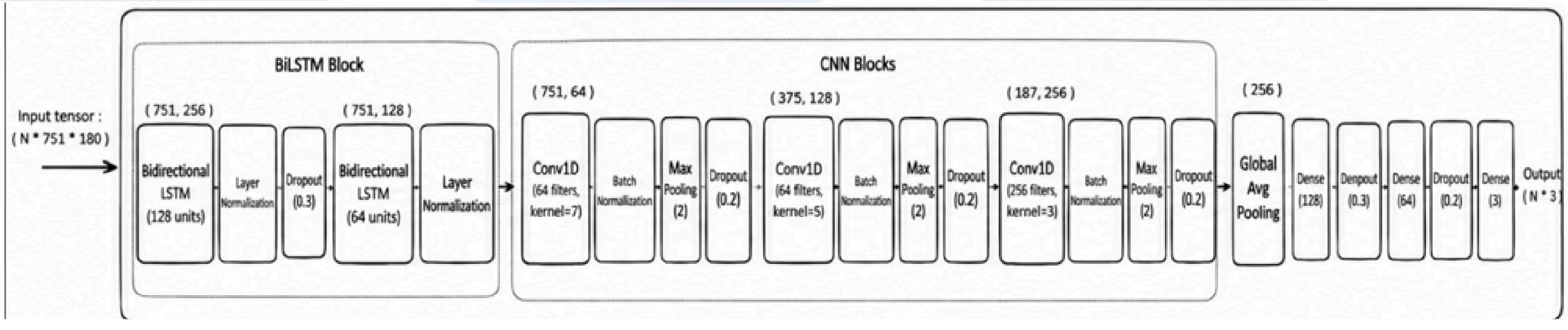
Logistic Regression

04 Evaluation

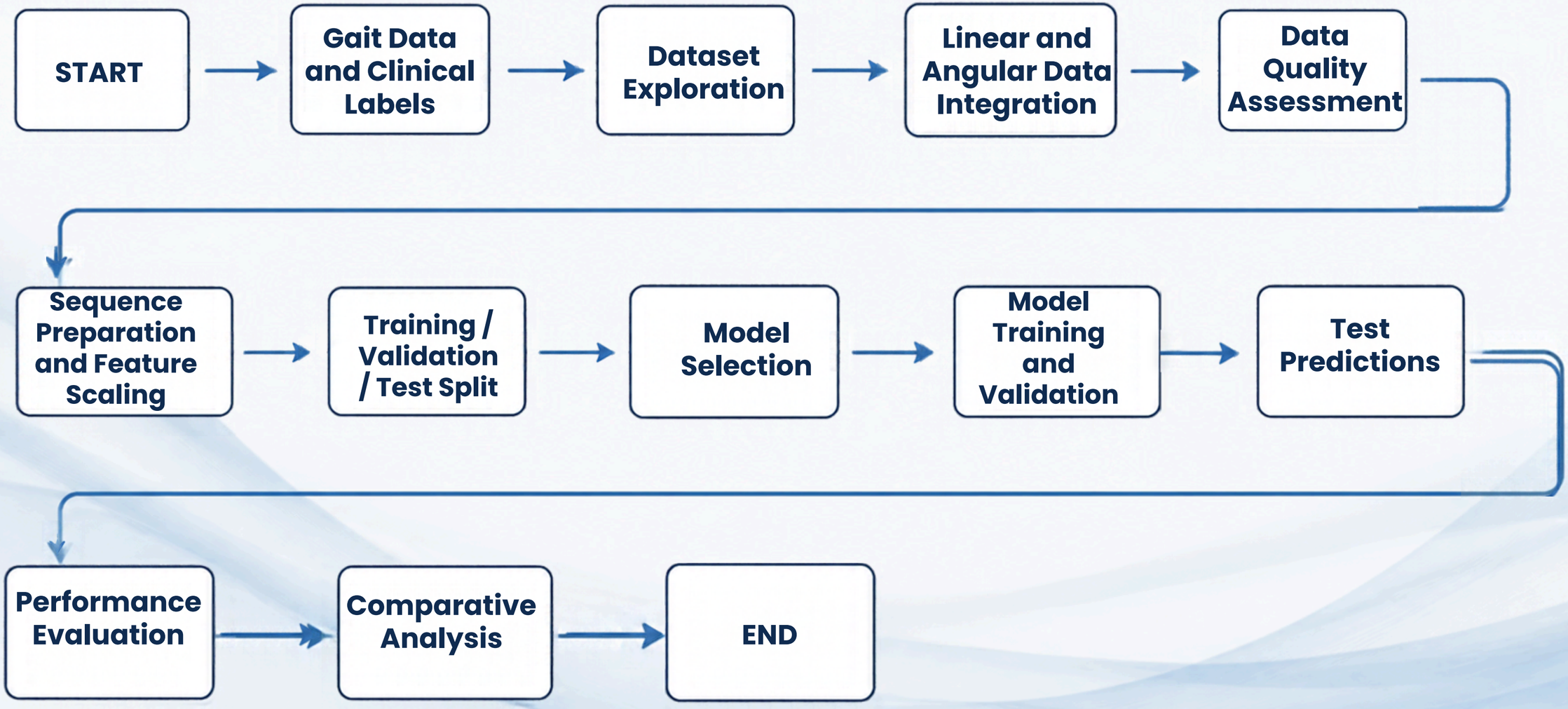
Accuracy

Balanced Accuracy

Macro F1-score



System Diagram / Workflow



Implementation

NOTEBOOKS	PHASE
1 - 2	Dataset exploration and understanding
3 - 5	Data collection,organization and preparation
6 - 7	Feature extraction and feature engineering
8	Data normalization and scaling
9	Sequence windowing and label preparation
10	Data quality and preprocessing validation
11	Baseline model development and performance evaluation

Python

Jupiter Notebook

PyTorch

Scikit-learn

NumPy

Pandas

CUDA

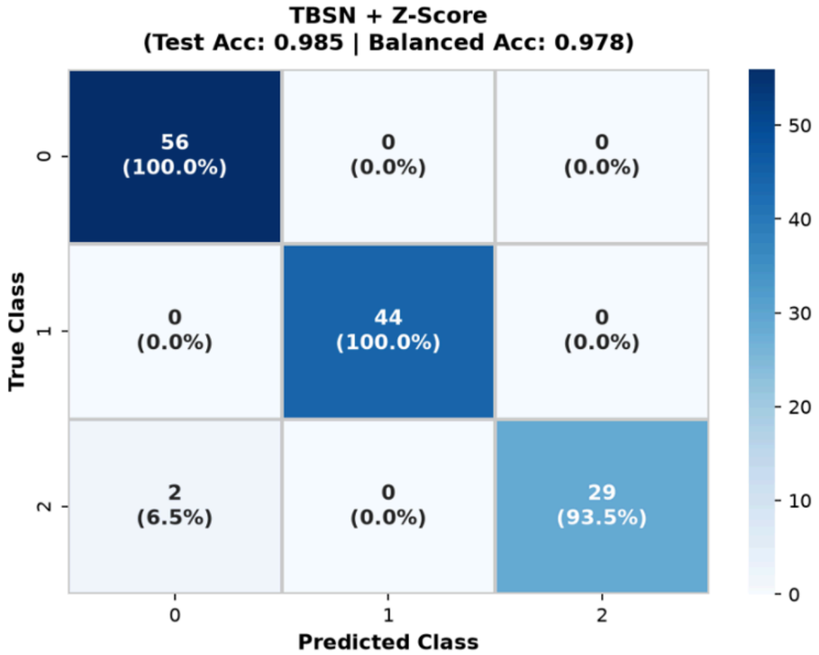


EXPERIMENTAL SETUP

Training trials	619
Validation trials	131
Testing trials	131
Total trials	881

Results, Comparison, and Analysis

Model	Accuracy	Balanced Accuracy	Macro F1
CNN1D	98.47%	97.85%	98.15%
LSTM	95.42%	94.18%	95.04%
BILSTM	98.47%	97.85%	98.30%
BILSTM-CNN	98.47%	97.89%	98.30%
Logistic Regression	98.47%	97.85%	98.15%



Classification Matrix of BiLSTM-CNN Model

Key Observations

98.47%
Maximum Accuracy

- Logistic Regression performed competitively.

97.89%
Maximum Balanced Accuracy

- BiLSTM-CNN achieved the highest balanced accuracy.

98.30%
Maximum macro F1

- BiLSTM and BiLSTM-CNN achieved the highest macro F1-score.

Conclusion and Future Work

Developed sensor-based gait-severity classification framework.

Compared five machine learning and deep learning models.

Achieved maximum test accuracy of 98.47%.

Observed influence of preprocessing choices on model performance.

Established a comparative experimental framework.

Future Work:

Model Interpretability

- Identify which kinematic features contribute most to gait-severity predictions.
- Use feature importance and explainability techniques to understand model decisions.

Advanced Model Architectures

- Implement attention-based models like transformer,BiLSTM with attention.

LIMITATION

Small Dataset Size

- The dataset contains only 26 participants and 881 gait trials.
- This limits the extent to which the findings can be generalized to a larger population.



References

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GitHUB Link : https://github.com/ATUL-IIITB/GROUP21_MINOR_PROJECT

References

Thank You .....